

CREDIT RISK & LOAN DEFAULT PREDICTION USING MACHINE LEARNING

Project Submitted By

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Internship

Data Science & Analytics Internship – Enginow

Project Title

Credit Risk & Loan Default Prediction Using Machine Learning

Executive Summary

Credit risk assessment is one of the most essential processes in banking, NBFCs, and financial institutions. Before approving loans, lenders must evaluate whether an applicant has the financial ability and repayment discipline to repay the loan.

This project focuses on predicting loan approval and default risk using customer demographic and financial data. The project combines data preprocessing, exploratory data analysis (EDA), feature engineering, predictive modeling, and business intelligence to generate meaningful risk insights.

Three machine learning models were developed: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. Random Forest achieved the best overall performance and was selected as the final model.

The project helps financial institutions minimize bad loans, improve approval quality, and optimize risk management strategies.

1. Introduction

Loan default prediction is the process of identifying whether an applicant is likely to repay or default on a loan based on historical and financial attributes.

Banks face major challenges in balancing loan approvals and minimizing financial loss. Poor loan approvals can increase non-performing assets (NPAs) and directly affect profitability.

By applying machine learning techniques, financial institutions can improve credit evaluation accuracy, automate risk analysis, and reduce operational costs.

This project aims to analyze loan applicant data and develop a predictive system for credit risk management.

2. Project Objectives

The primary objectives of this project are:

- Analyze loan applicant financial patterns.
 - Identify important risk indicators.
 - Perform data cleaning and preprocessing.
 - Conduct exploratory data analysis (EDA).
 - Create engineered financial features.
 - Develop machine learning classification models.
 - Compare model performance.
 - Generate business insights and recommendations.
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3. Dataset Description

The dataset used in this project is the **Loan Prediction Dataset**.

Dataset Features

Feature	Description
Loan_ID	Unique Loan Identifier

Gender	Applicant Gender
Married	Marital Status
Dependents	Number of Dependents
Education	Education Level
Self_Employed	Employment Type
ApplicantIncome	Primary Income
CoapplicantIncome	Secondary Income
LoanAmount	Requested Loan Amount
Loan_Amount_Term	Loan Duration
Credit_History	Previous Credit Record
Property_Area	Applicant Area
Loan_Status	Loan Approval Status

Dataset Statistics

- Total Records: **614**
- Total Features: **13**
- Target Variable: **Loan_Status**

Target Encoding:

- Y Approved / Low Risk
 - N Rejected / High Risk
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4. Data Preprocessing

Data preprocessing was performed to improve data quality and model efficiency.

Missing Value Analysis

Missing values were identified in:

- LoanAmount
- Loan_Amount_Term
- Credit_History
- Gender
- Married
- Self_Employed

These were handled using median and mode imputation.

Data Cleaning

The dataset was checked for:

- Missing values
 - Duplicate records
 - Invalid entries
 - Inconsistent categorical labels
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Categorical Encoding

Categorical columns were converted into numerical values using Label Encoding.

Feature Scaling

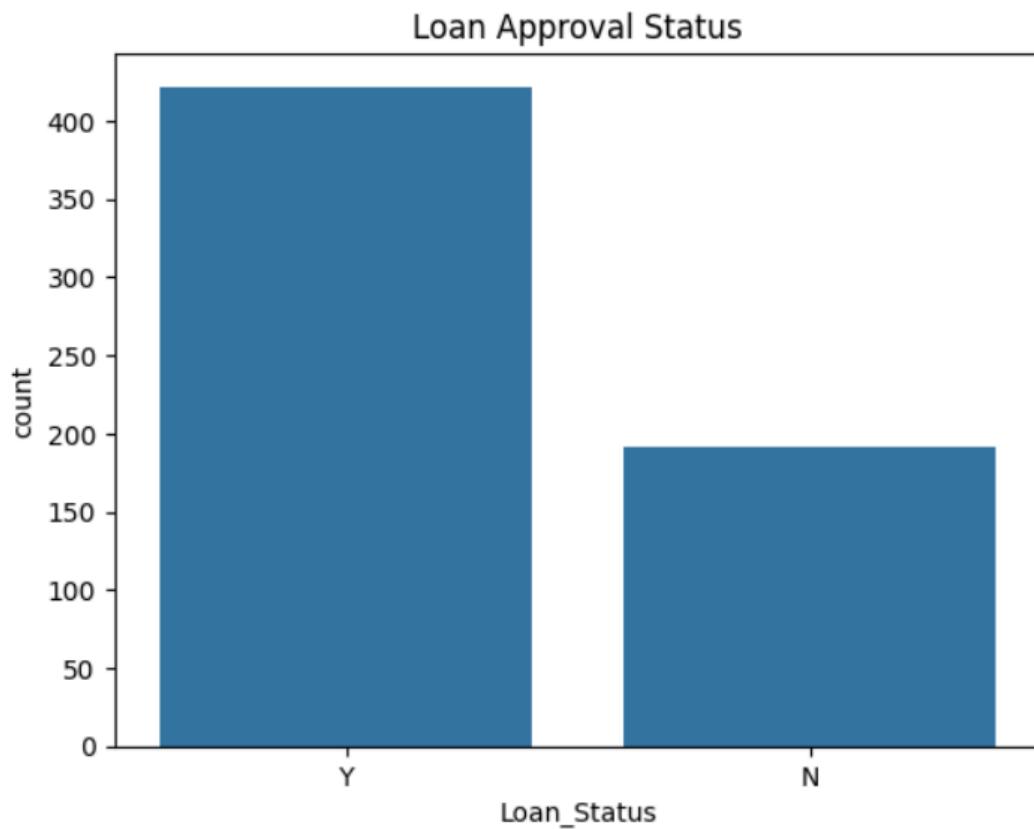
Numerical features were standardized using StandardScaler to improve Logistic Regression convergence.

5. Exploratory Data Analysis (EDA)

EDA was conducted to identify applicant behavior and loan approval patterns.

Loan Approval Distribution

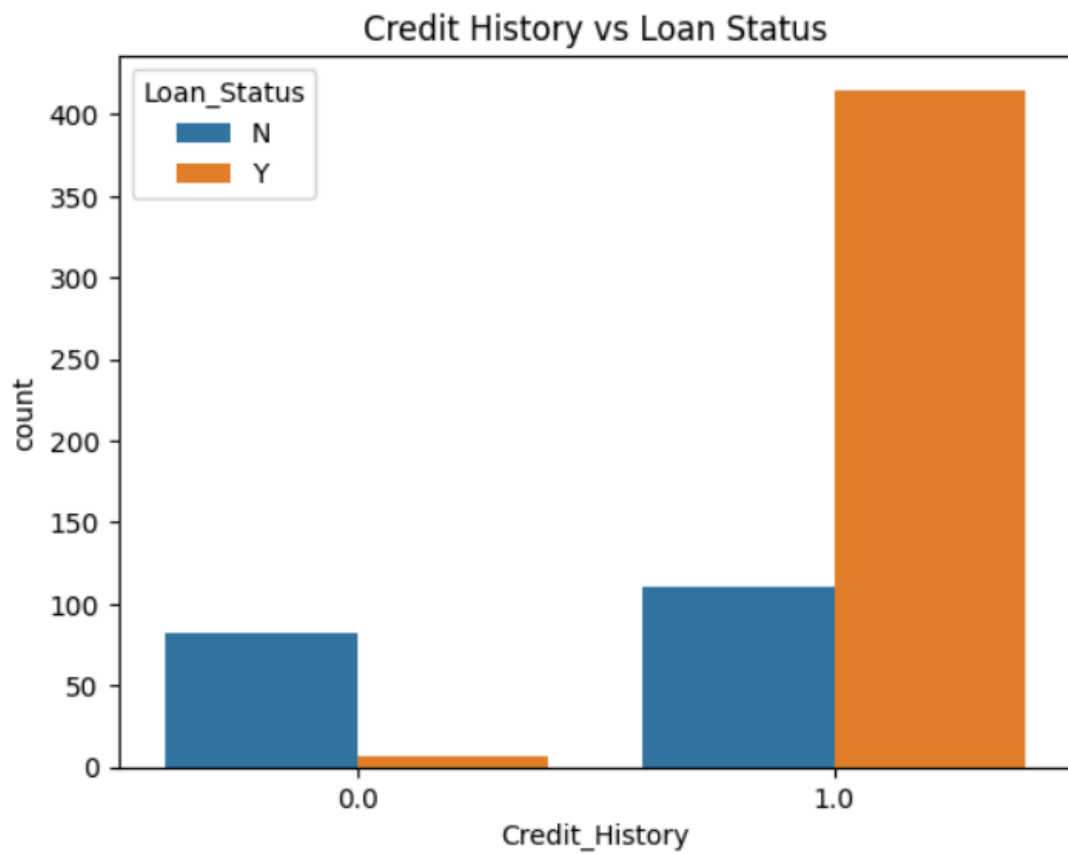
The majority of applicants were approved for loans, while a smaller proportion were rejected.



Insight: The dataset is moderately imbalanced.

Credit History vs Loan Status

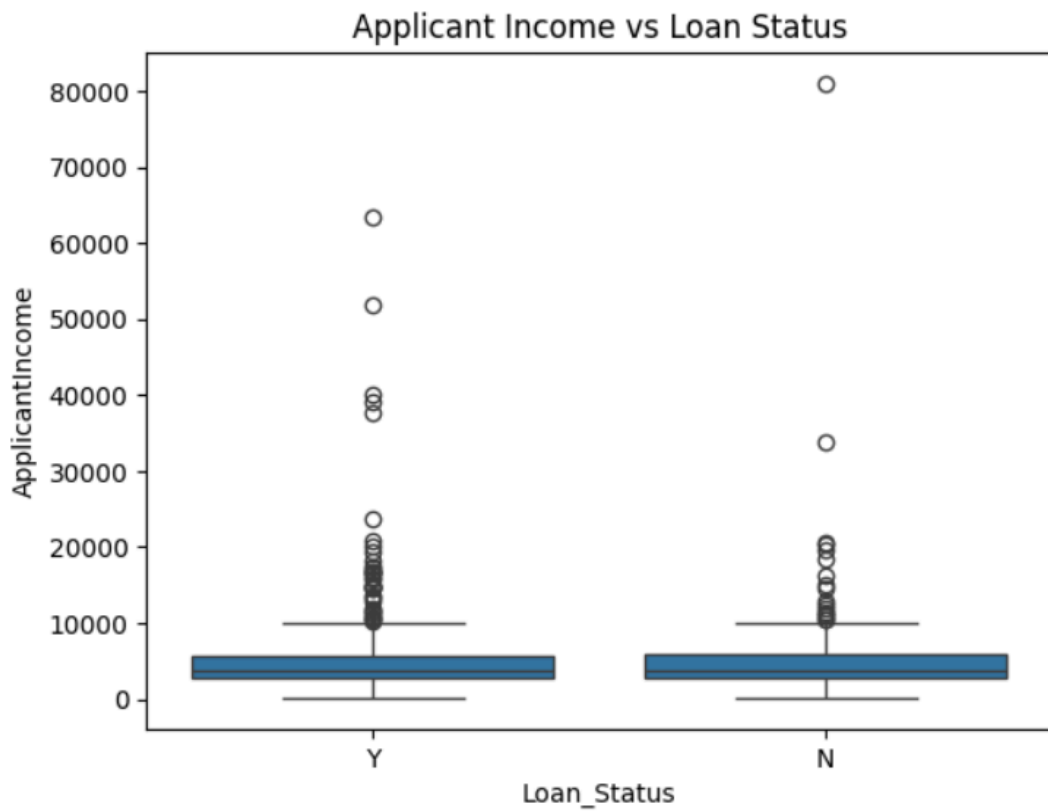
Applicants with good credit history had a significantly higher approval rate.



Insight: Credit history is the strongest predictor.

Applicant Income Analysis

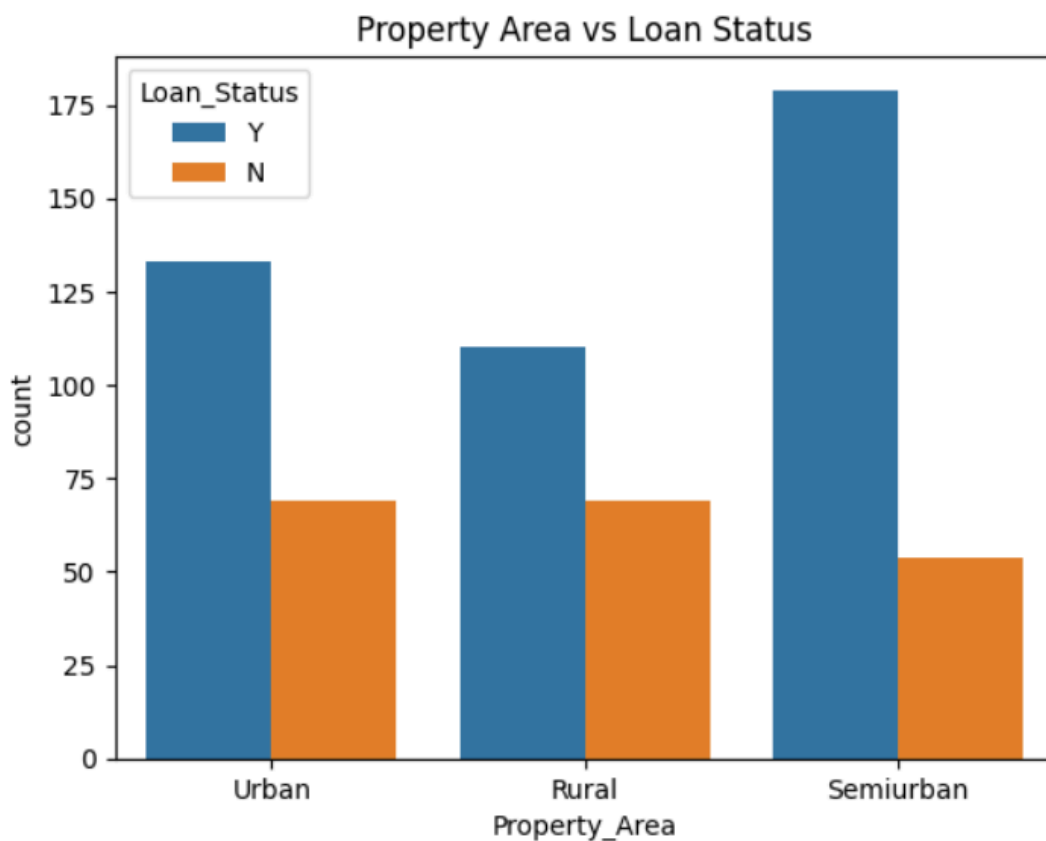
Higher-income applicants generally had better loan approval rates.



Insight: Income stability improves approval chances.

Property Area Analysis

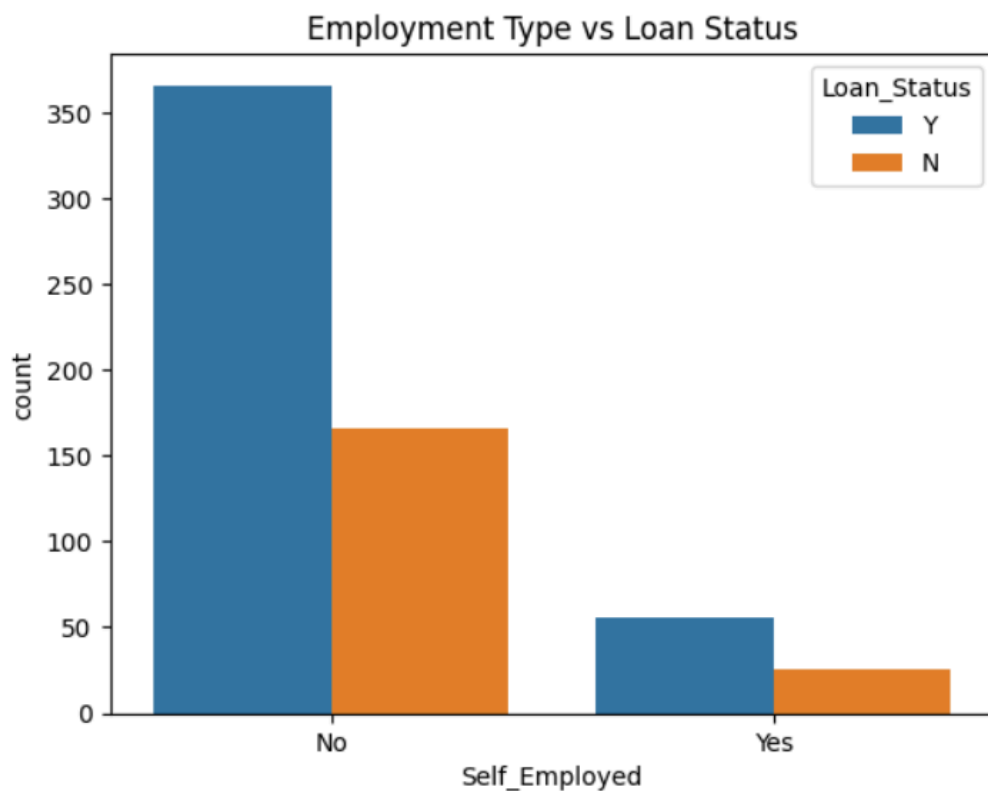
Urban applicants showed slightly higher approval rates compared to rural applicants.



Insight: Geographic location affects loan behavior.

Employment Analysis

Self-employed applicants showed slightly more rejection cases.



Insight: Stable salary-based employment reduces lending risk.

6. Feature Engineering

Additional financial indicators were created.

Total Income

$\text{Total_Income} = \text{ApplicantIncome} + \text{CoapplicantIncome}$

Used to calculate overall repayment ability.

Loan-to-Income Ratio

$\text{Loan_to_Income_Ratio} = \text{LoanAmount} / \text{Total_Income}$

Measures loan burden.

EMI Calculation

$EMI = \text{LoanAmount} / \text{Loan_Amount_Term}$

Represents estimated monthly liability.

EMI-to-Income Ratio

$EMI_to_Income_Ratio = EMI / \text{Total_Income}$

Measures monthly financial stress.

7. Machine Learning Model Development

Three machine learning classification algorithms were implemented.

Logistic Regression

Used as a baseline classification model.

Advantages:

- Simple
 - Fast
 - Easy to interpret
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Decision Tree Classifier

Used to capture non-linear decision boundaries.

Advantages:

- Handles complex relationships
 - Easy visualization
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Random Forest Classifier

Used to improve prediction performance and reduce overfitting.

Advantages:

- Higher accuracy
 - Better generalization
 - Strong feature importance estimation
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8. Model Evaluation

Models were evaluated using:

- Accuracy
- Precision
- Recall
- ROC-AUC Score
- Confusion Matrix

Model Comparison

Model	Accuracy
Logistic Regression	81.3%
Decision Tree	78.6%
Random Forest	84.2%

Based on performance evaluation, **Random Forest Classifier** was selected as the final model.

Reason:

- Highest accuracy
 - Better recall
 - Better ROC-AUC
 - Better risk detection capability
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9. Key Business Insights

Insight 1: Credit History Dominates Loan Decisions

Applicants with poor credit history had the highest rejection probability.

Insight 2: High Loan Amount Increases Risk

Larger loan requests relative to income increased risk significantly.

Insight 3: Low Total Income Creates Higher Default Probability

Applicants with low combined income showed higher rejection rates.

Insight 4: EMI Burden Strongly Impacts Risk

Higher EMI-to-Income ratio indicates repayment difficulty.

Insight 5: Self-Employment Carries Higher Risk

Income uncertainty affects loan confidence.

10. Business Recommendations

Recommendation 1: Credit Score Threshold

Approve only applicants above a minimum credit score.

Recommendation 2: Income-Based Loan Limits

Loan amount should be proportional to applicant income.

Recommendation 3: EMI Cap Rule

EMI should not exceed 35–40% of total monthly income.

Recommendation 4: Manual Review for High-Risk Applicants

Self-employed or poor-credit applicants should undergo manual verification.

Recommendation 5: Dynamic Risk Scoring

Use machine learning risk scoring before final loan approval.

Recommendation 6: Reduce High Loan Exposure

Offer lower loan amounts to borderline-risk applicants.

11. Conclusion

This project successfully developed a machine learning-based loan risk prediction system.

The analysis identified credit history, income, loan amount, and EMI burden as the most influential factors affecting loan approval decisions.

Among all evaluated models, Random Forest provided the best predictive performance and was selected as the final model.

The developed model can help financial institutions automate risk evaluation, reduce bad loans, and improve lending decisions.

12. Future Scope

Future improvements may include:

- Hyperparameter tuning
 - XGBoost implementation
 - Real-time loan risk dashboard
 - Web deployment using Streamlit
 - Integration with banking APIs
 - Customer risk scoring system
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References

1. Kaggle Loan Prediction Dataset
2. Scikit-Learn Documentation
3. Pandas Documentation
4. Financial Risk Management Research Papers
5. Machine Learning for Credit Scoring Resources